Python vs. Traditional Software

GeoPython — Quick Reference Handout 02

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1 The Core Difference: Reproducibility

Packaged software (Graphical User Interface, or GUI-based tools) is excellent for producing publication-ready, polished single graphs. Python is used for a fundamentally different reason: reproducibility. When your data updates tomorrow, you run the script again — no manual re-plotting required.

Dimension	Packaged Software (GUI)	Python Script
Workflow	Click, drag, format manually	Write once, re-run anytime
Data update	Redo steps from scratch	Run → output refreshes automatically
Collaboration	Share file + instructions	Share script — it IS the instructions
Scale	One file at a time	Loops over hundreds of files
Auditability	Hard to trace decisions	Every step is written in plain code

2 When to Use Each Tool

There is no "winner." Use the right tool for the job:

Use Case	Best Tool
Final figure for a journal paper (one-time polish)	Packaged Software (GUI)
Processing 50 field-season CSV files the same way	Python
Quick exploratory chart to check your data	Either — Python is faster at scale
Automated report sent monthly to a stakeholder	Python
Collaborative analysis others must verify step-by-step	Python

3 The Reproducibility Mindset

Reproducibility is not about being "more technical." It is about saving yourself time and building trust in your results.

! *Every manual step you take in a Graphical User Interface (GUI) is a step someone else (or future-you) cannot verify. A Python script is self-documenting: it shows exactly what happened to the data.*

If you would be uncomfortable describing your analysis in writing, that is a signal to move it into a script. The code is the methods section.